

Electronic Devices

Final Term Lecture - 02

Reference book:

Electronic Devices and Circuit Theory (Chapter-5)

Robert L. Boylestad and L. Nashelsky , (11th Edition)



Faculty of Engineering

American International University-Bangladesh

COMMON-EMITTER VOLTAGE-DIVIDER BIAS

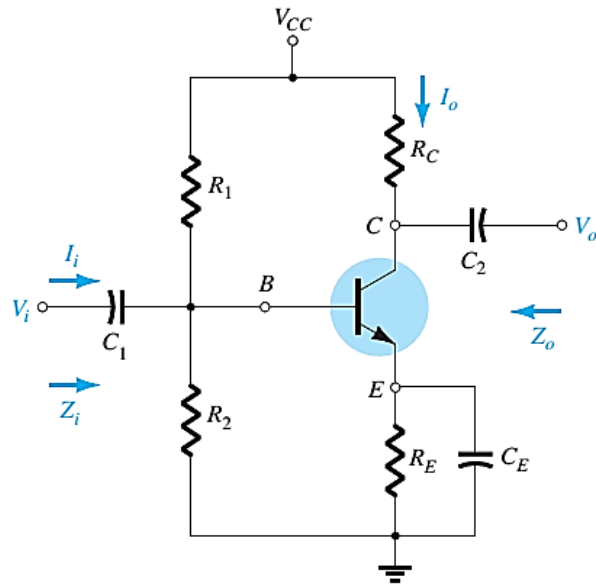


FIG. 5.26

Voltage-divider bias configuration.

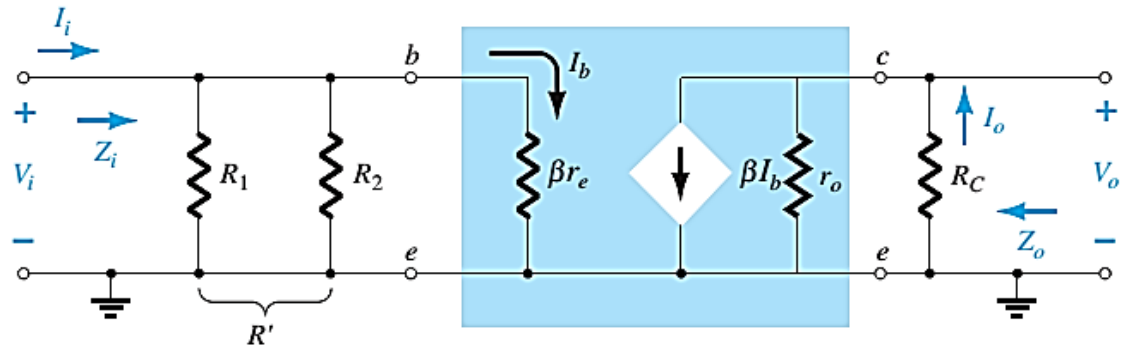


FIG. 5.27

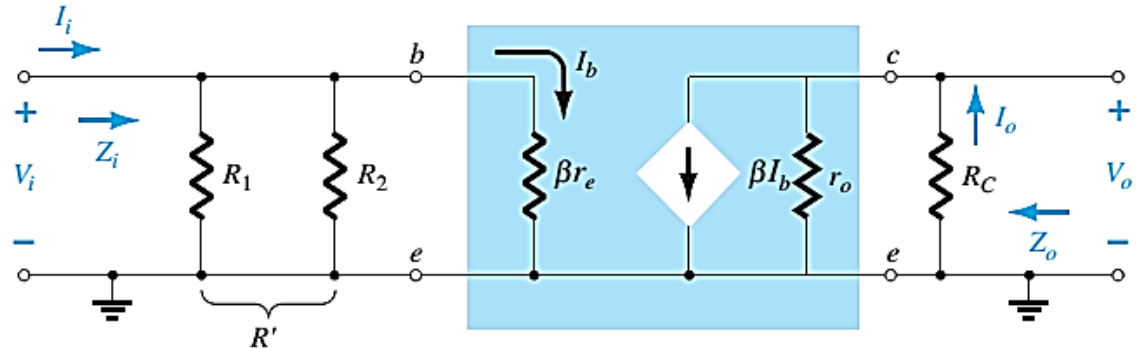
Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 5.26.

COMMON-EMITTER VOLTAGE-DIVIDER BIAS

INPUT IMPEDANCE, Z_i

$$R' = R_1 \parallel R_2$$

$$Z_i = R' \parallel \beta r_e$$



OUTPUT IMPEDANCE, Z_o

$$Z_o = R_C \parallel r_o$$

$$Z_o \cong R_C \mid_{r_o \geq 10R_C}$$

VOLTAGE GAIN, A_v

$$V_o = -\beta I_b (R_C \parallel r_o) = -\beta \left(\frac{V_i}{\beta r_e} \right) (R_C \parallel r_o); I_b = \frac{V_i}{\beta r_e}$$

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \parallel r_o)}{r_e}, A_v = -\frac{R_C}{r_e} \mid_{r_o \geq 10R_C}$$



EXAMPLE

- **EXAMPLE 5.2:** For the network of Fig. 5.28 :
- Determine r_e , Z_i , Z_o (with $r_o = \infty$), A_v (with $r_o = \infty$) and Repeat with $r_o = 50 \text{ k}\Omega$.

a. DC: Testing $\beta R_E > 10R_2$,

$$(90)(1.5 \text{ k}\Omega) > 10(8.2 \text{ k}\Omega)$$

$$135 \text{ k}\Omega > 82 \text{ k}\Omega \text{ (satisfied)}$$

Using the approximate approach, we obtain

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{(8.2 \text{ k}\Omega)(22 \text{ V})}{56 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 2.81 \text{ V}$$

$$V_E = V_B - V_{BE} = 2.81 \text{ V} - 0.7 \text{ V} = 2.11 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.11 \text{ V}}{1.5 \text{ k}\Omega} = 1.41 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.41 \text{ mA}} = \mathbf{18.44 \text{ }\Omega}$$

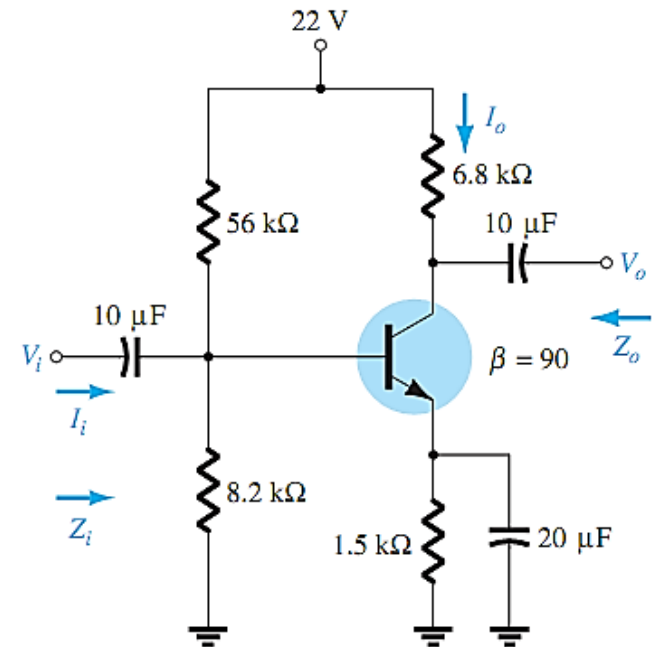
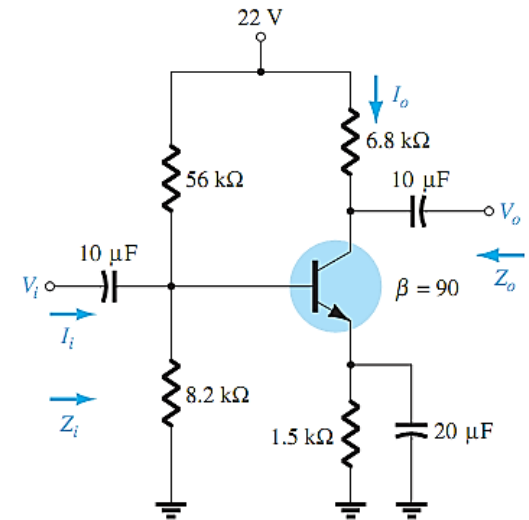


FIG. 5.28

Example 5.2.

EXAMPLE Contd.

- b. $R' = R_1 \parallel R_2 = (56 \text{ k}\Omega) \parallel (8.2 \text{ k}\Omega) = 7.15 \text{ k}\Omega$
 $Z_i = R' \parallel \beta r_e = 7.15 \text{ k}\Omega \parallel (90)(18.44 \text{ }\Omega) = 7.15 \text{ k}\Omega \parallel 1.66 \text{ k}\Omega$
 $= 1.35 \text{ k}\Omega$
- c. $Z_o = R_C = 6.8 \text{ k}\Omega$
- d. $A_v = -\frac{R_C}{r_e} = -\frac{6.8 \text{ k}\Omega}{18.44 \text{ }\Omega} = -368.76$
- e. $Z_i = 1.35 \text{ k}\Omega$
 $Z_o = R_C \parallel r_o = 6.8 \text{ k}\Omega \parallel 50 \text{ k}\Omega = 5.98 \text{ k}\Omega$ vs. $6.8 \text{ k}\Omega$
 $A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{5.98 \text{ k}\Omega}{18.44 \text{ }\Omega} = -324.3$ vs. -368.76



There was a measurable difference in the results for Z_o and A_v , because the condition $r_o \geq 10R_C$ was *not* satisfied.



COMMON-EMITTER EMITTER-BIAS CONFIGURATION: UNBYPASSED R_E

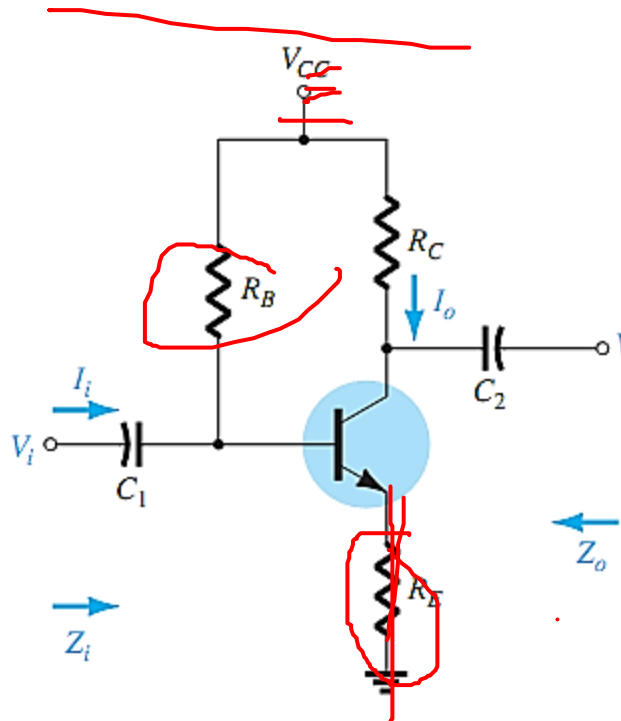


FIG. 5.29

CE emitter-bias configuration.

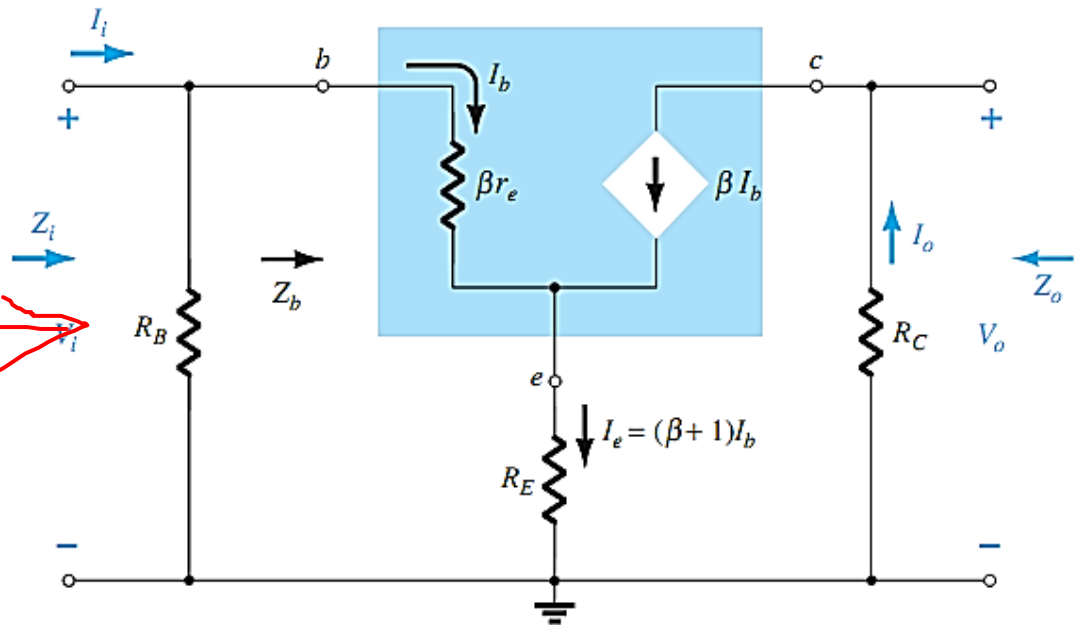


FIG. 5.30

Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 5.29.

IMPEDANCE CALCULATION

INPUT IMPEDANCE, Z_i

$$V_i = I_b \beta r_e + I_e R_E$$

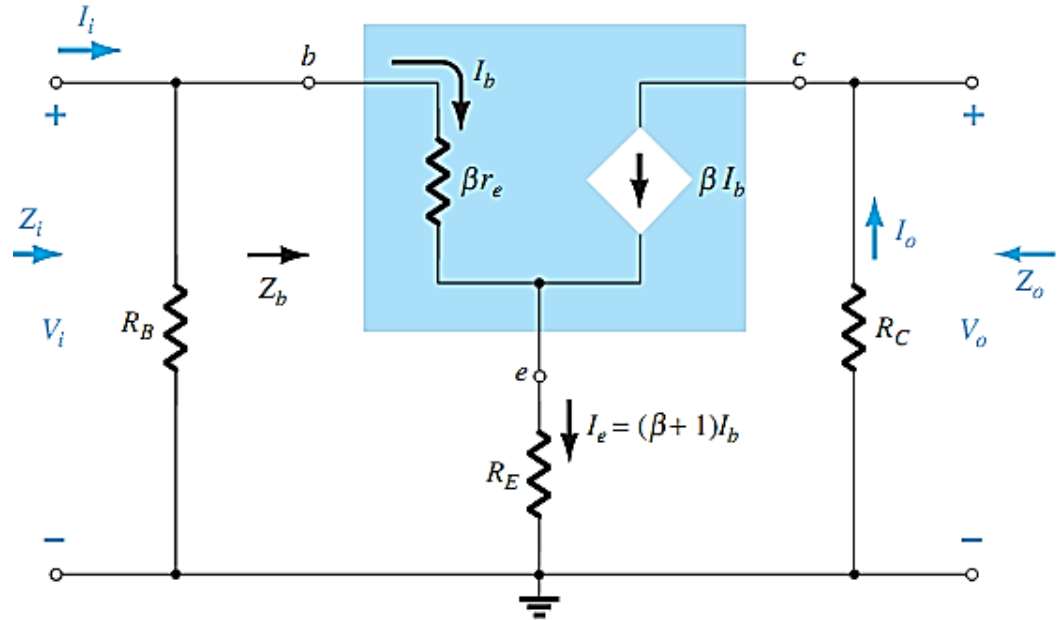
$$= I_b \beta r_e + (\beta + 1) I_b R_E$$

$$Z_b = \frac{V_i}{I_b} = \beta r_e + (\beta + 1) R_E$$

$$Z_b \cong \beta r_e + \beta R_E = \beta(r_e + R_E)$$

$$Z_b \cong \beta R_E \quad \text{for } R_E \gg r_e$$

$$Z_i = R_B || Z_b$$



OUTPUT IMPEDANCE, Z_o

$$Z_o = R_C$$

GAIN CALCULATIONS

VOLTAGE GAIN, A_v

$$V_o = -I_o R_C = -\beta I_b R_C = -\beta \left(\frac{V_i}{Z_b} \right) R_C$$

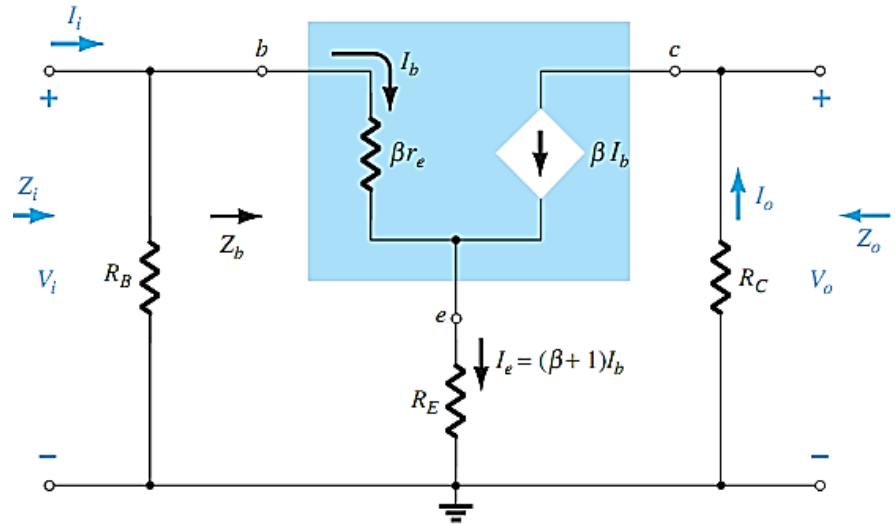
$$A_v = \frac{V_o}{V_i} = -\frac{\beta R_C}{Z_b}$$

Substituting $Z_b \cong \beta(r_e + R_E)$

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{r_e + R_E}$$

For the approximation $Z_b \cong \beta R_E$

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{R_E}$$



The **negative sign** in gain equations reveals **180° phase shift** between input and output waveforms.

COMMON-EMITTER EMITTER-BIAS CONFIGURATION: BYPASSED R_E

Bypassed

If R_E is bypassed by an emitter capacitor C_E , the complete r_e equivalent model can be substituted, resulting in the same equivalent network as Fig. 5.22. Equations of slide no. 13 are therefore applicable.

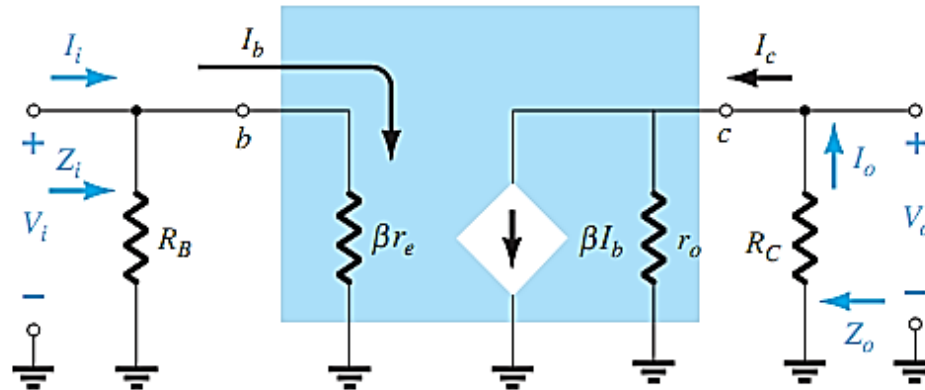


FIG. 5.22

Substituting the r_e model into the network of Fig. 5.21.

EXAMPLE

- EXAMPLE 5.3:** For the network of following Fig, without C_E (unbypassed), determine: r_e , Z_i , Z_o & A_v .

a. DC:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (121)0.56 \text{ k}\Omega} = 35.89 \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (121)(35.89 \mu\text{A}) = 4.34 \text{ mA}$$

and $r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{4.34 \text{ mA}} = \mathbf{5.99 \Omega}$

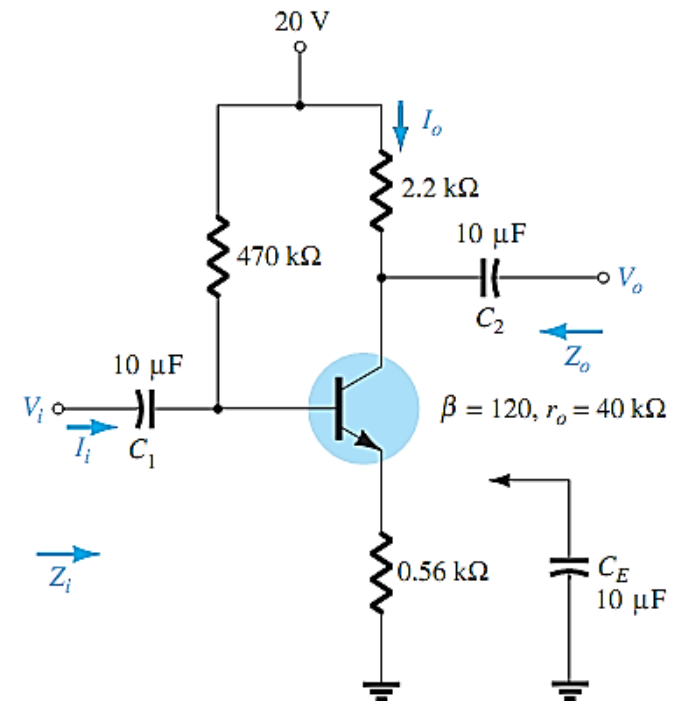


FIG. 5.32
Example 5.3.

EXAMPLE Contd.

b. Testing the condition $r_o \geq 10(R_C + R_E)$, we obtain

$$40 \text{ k}\Omega \geq 10(2.2 \text{ k}\Omega + 0.56 \text{ k}\Omega)$$

$$40 \text{ k}\Omega \geq 10(2.76 \text{ k}\Omega) = 27.6 \text{ k}\Omega \text{ (satisfied)}$$

Therefore,

$$\begin{aligned} Z_b &\cong \beta(r_e + R_E) = 120(5.99 \Omega + 560 \Omega) \\ &= 67.92 \text{ k}\Omega \end{aligned}$$

and

$$\begin{aligned} Z_i &= R_B \parallel Z_b = 470 \text{ k}\Omega \parallel 67.92 \text{ k}\Omega \\ &= \mathbf{59.34 \text{ k}\Omega} \end{aligned}$$

c. $Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$

d. $r_o \geq 10R_C$ is satisfied. Therefore,

$$\begin{aligned} A_v &= \frac{V_o}{V_i} \cong -\frac{\beta R_C}{Z_b} = -\frac{(120)(2.2 \text{ k}\Omega)}{67.92 \text{ k}\Omega} \\ &= \mathbf{-3.89} \end{aligned}$$

compared to -3.93 using Eq. (5.20): $A_v \cong -R_C/R_E$.

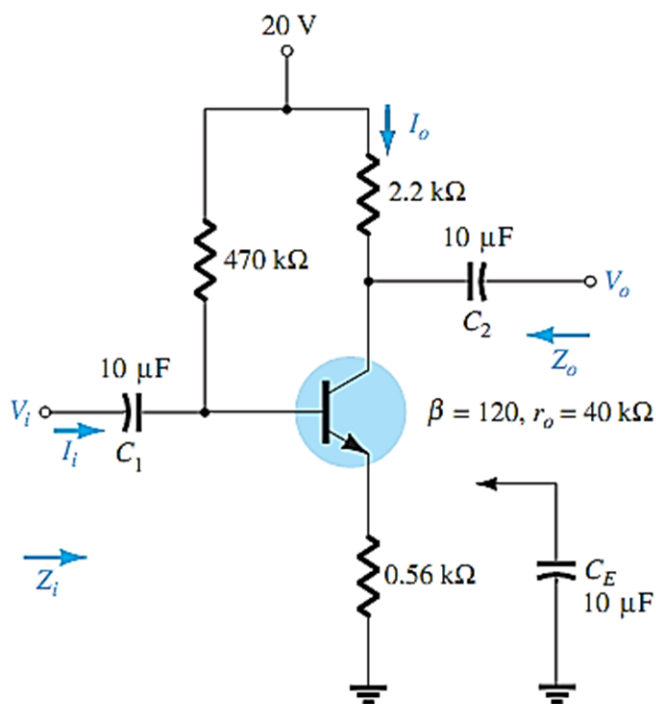


FIG. 5.32

Example 5.3.

EMITTER-FOLLOWER CONFIGURATION

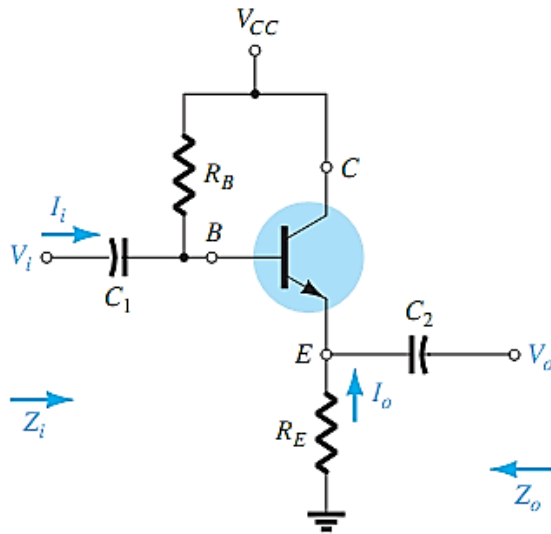


FIG. 5.36

Emitter-follower configuration.

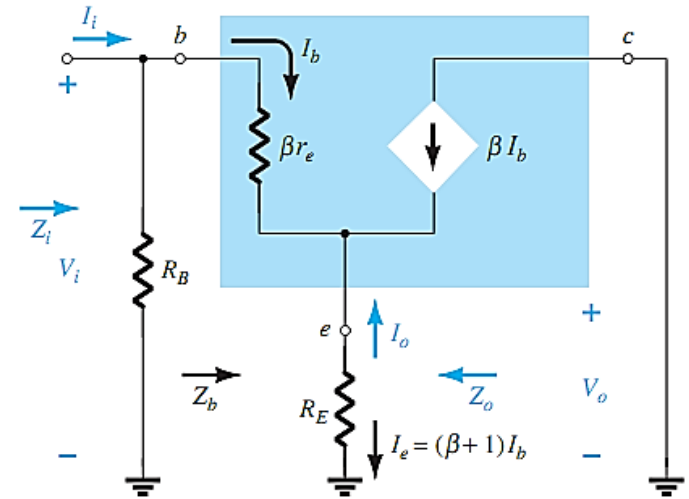


FIG. 5.37

Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 5.36.

- This is also known as the common-collector configuration.
- The input is applied to the base and the output is taken from the emitter.
- There is no phase shift between input and output.

IMPEDANCE CALCULATIONS

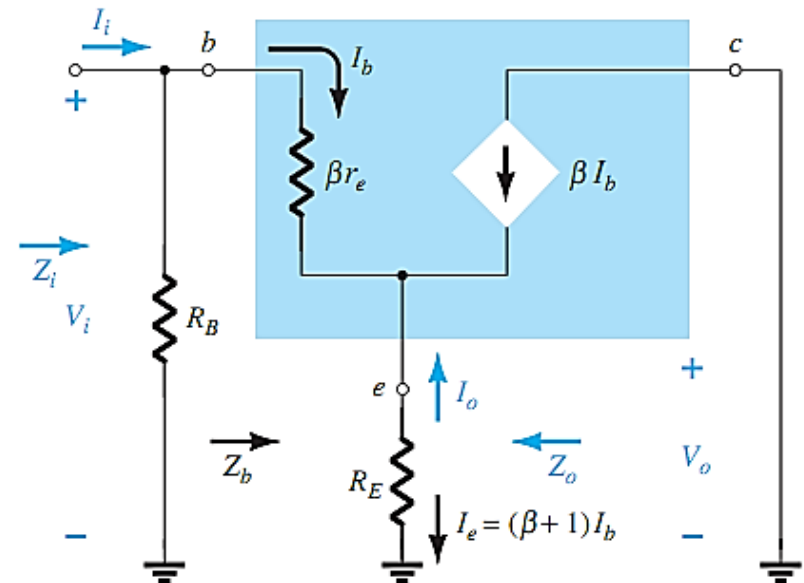
INPUT IMPEDANCE, Z_i

$$Z_i = R_B \parallel Z_b$$

$$Z_b = \beta r_e + (\beta + 1)R_E$$

$$Z_b \cong \beta(r_e + R_E)$$

$$Z_b \cong \beta R_E \quad \text{for } R_E \gg r_e$$



IMPEDANCE CALCULATIONS

OUTPUT IMPEDANCE, Z_o

$$I_b = \frac{V_i}{Z_b},$$

$$I_E = (\beta + 1)I_b = (\beta + 1) \frac{V_i}{Z_b}$$

$$I_E = \frac{(\beta + 1)V_i}{\beta r_e + (\beta + 1)R_E}$$

Since $(\beta + 1) \cong \beta$

$$I_E = \frac{V_i}{r_e + R_E}$$

To determine Z_o , V_i is set to zero

$$Z_o = R_E || r_e \quad Z_o = r_e \mid_{R_E \gg r_e}$$

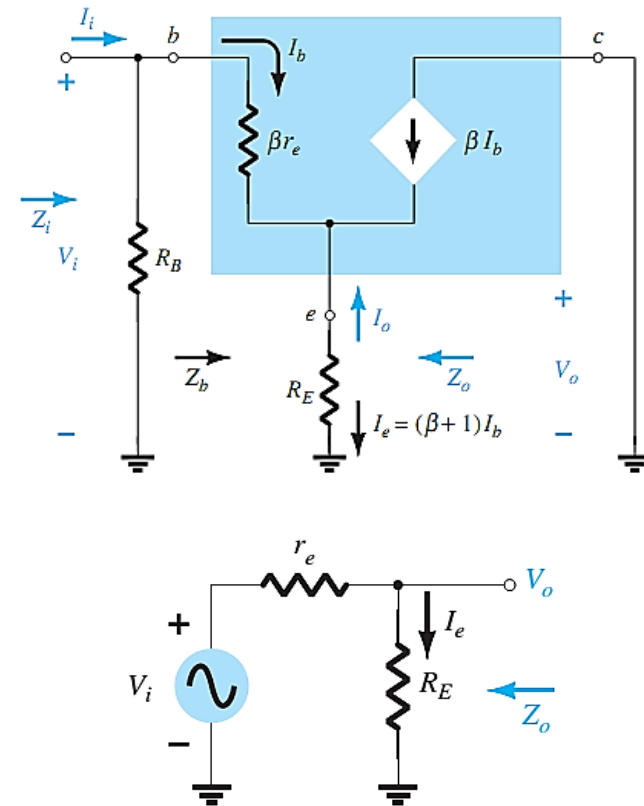


FIG. 5.38

Defining the output impedance for the emitter-follower configuration.

GAIN CALCULATIONS

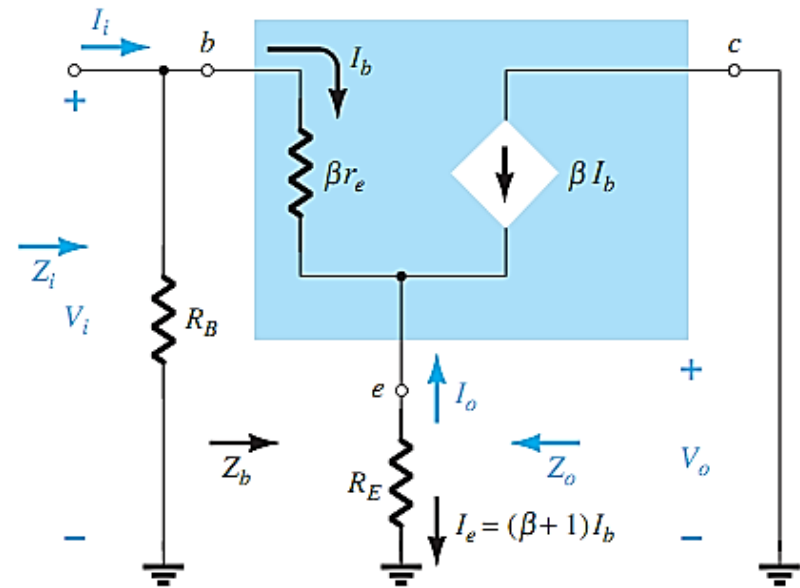
VOLTAGE GAIN, A_v

$$V_o = I_e R_E = (\beta + 1) I_b R_E = \frac{V_i (\beta + 1) R_E}{Z_b}$$

when $r_o \geq 10 R_E$ and $\beta + 1 \cong \beta$

But $Z_b \cong \beta(r_e + R_E)$

$$A_v = \frac{V_o}{V_i} \cong \frac{\beta R_E}{\beta(r_e + R_E)} \cong \frac{R_E}{(r_e + R_E)}$$



□ See Example 5.7

COMMON-BASE CONFIGURATION

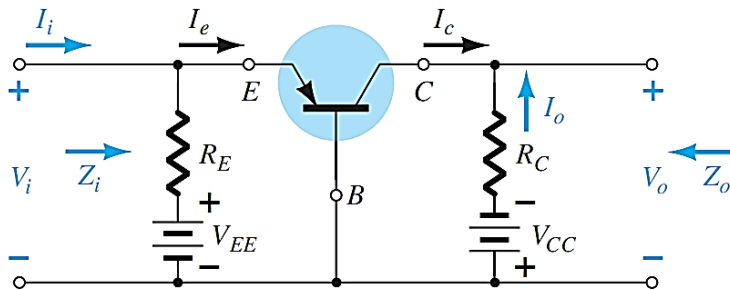


FIG. 5.42

Common-base configuration.

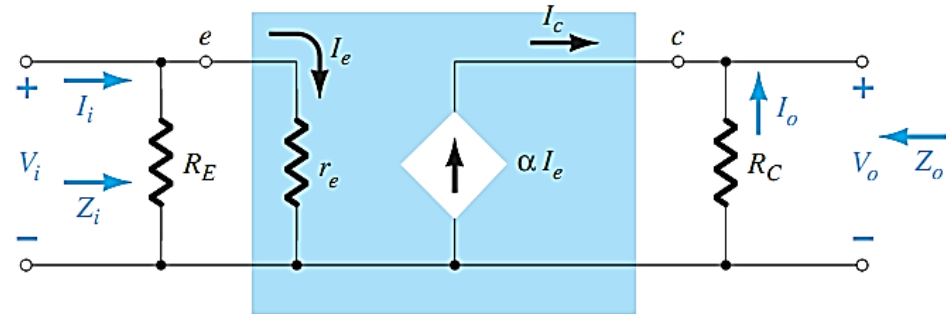


FIG. 5.43

Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 5.44.

- The input is applied to the emitter.
- The output is taken from the collector.
- Low input impedance.
- High output impedance.
- Very high voltage gain.
- No phase shift between input and output.

CALCULATIONS

INPUT IMPEDANCE, Z_i

$$Z_i = R_E \parallel r_e$$

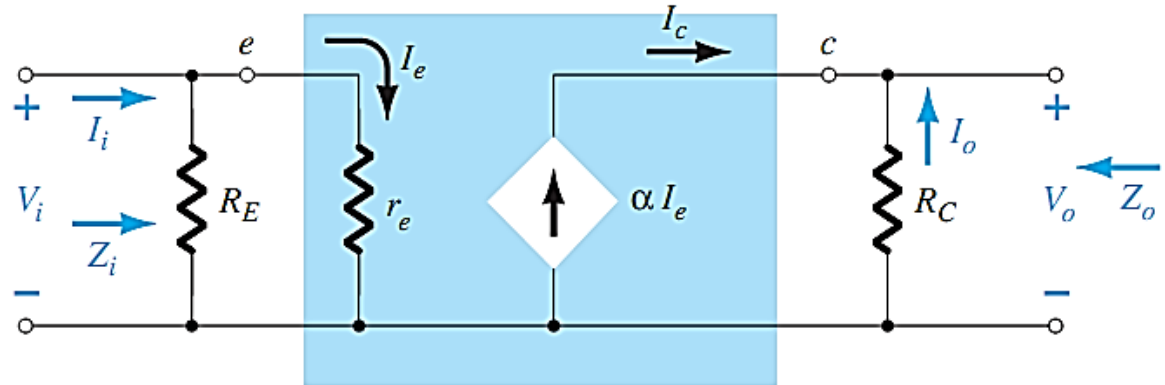
OUTPUT IMPEDANCE, Z_o

$$Z_o = R_C$$

VOLTAGE GAIN, A_v

$$V_o = -I_o R_C = -(I_C) R_C = \alpha I_e R_C; \quad I_e = \frac{V_i}{r_e}$$

$$V_o = \alpha \left(\frac{V_i}{r_e} \right) R_C; \quad A_v = \frac{V_o}{V_i} = \frac{\alpha R_C}{r_e} \cong \frac{R_C}{r_e}$$



CURRENT GAIN, A_i

Assuming $R_E \gg r_e$

$$I_e = I_i$$

$$I_o = -\alpha I_e = -\alpha I_i$$

$$A_i = \frac{I_o}{I_i} = -\alpha \cong -1$$

COMMON-BASE CONFIGURATION

- **Phase Relationship:**

The fact that A_v is a positive number shows that V_o and V_i are in phase for the common-base configuration.

- **Effect of r_o :**

For the common-base configuration, $r_o = 1/h_{ob}$ is typically in the megohm range and sufficiently larger than the parallel resistance R_C to permit the approximation $r_o \parallel R_C \cong R_C$.



EXAMPLE

- EXAMPLE 5.8:** For the network of following figure, determine: r_e , Z_i , Z_o , A_v , A_i .

$$\text{a. } I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{2 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \frac{1.3 \text{ V}}{1 \text{ k}\Omega} = 1.3 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.3 \text{ mA}} = \mathbf{20 \Omega}$$

$$\text{b. } Z_i = R_E \parallel r_e = 1 \text{ k}\Omega \parallel 20 \Omega \\ = \mathbf{19.61 \Omega \cong r_e}$$

$$\text{c. } Z_o = R_C = \mathbf{5 \text{ k}\Omega}$$

$$\text{d. } A_v \cong \frac{R_C}{r_e} = \frac{5 \text{ k}\Omega}{20 \Omega} = \mathbf{250}$$

$$\text{e. } A_i = \mathbf{-0.98 \cong -1}$$

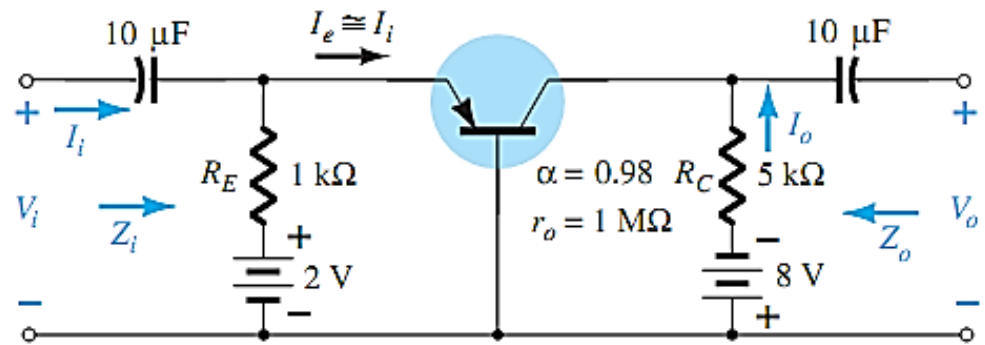


FIG. 5.44

Example 5.8.

End of Lecture-2

